Fall 2026 Math 110 Midterm 2 Review

Definitions and Theorems

Definition 1 (Quotient Space)**.** Given a subspace U of V we can define sets that are called **translations of U** via $v+U := \{v+u | u \in U\}$. Intuitively we think of $v+U$ as 'U shifted away from 0 by v', like lifting up a table. We define the **quotient space of V by U** denoted $V/U := \{v+U | v \in V\}$ to be the set of all translations of U. Addition of translations is defined by $(v_1+U)+(v_2+U) := (v_1+v_2)+U$ and so on and so forth. Together this set forms a vector space.

Theorem 1 (Quotient Map and Dimension)**.** We can define a linear map $\pi : V \to V/U$ from V to its quotient by any particular subspace U via $\pi(v) = v+U$. Note that this map is surjective, since for any $v+U$ its mapped onto by v. Moreover, the null space is identically U. If V is finite dimensional then by the Fundamental Theorem of Linear Maps we have

$$\dim V = \dim \operatorname{Null} \pi + \dim \operatorname{Range} \pi = \dim U + \dim V/U \implies \dim V/U = \dim V - \dim U$$

Theorem 2 (First Isomorphism Theorem)**.** For any linear map $T : V \to W$ we get a new map

$$\widetilde{T} : V/\operatorname{Null} T \to \operatorname{Range} T \text{ defined by } \widetilde{T}(v + \operatorname{Null} T) = T(v)$$

It's not obvious that this is even well-defined, but it is, and it's an *isomorphism*.

Definition 2 (Dual Space)**.** For any vector space V we can consider the vector space $\mathcal{L}(V, \mathbb{F})$ of linear maps from the vector space to its field. This space is called the **dual space of V** and is usually denoted V' (or sometimes V^*). If V is finite dimensional then $\dim V' = \dim V$.

Definition 3 (Dual Basis to a Basis)**.** For any basis $\beta = \{v_1, \ldots, v_n\}$ of vector space V we can define what is called its **dual basis** $\beta' = \{\varphi_1, \ldots, \varphi_n\}$ with each φ_i defined by its action on the basis β via $\varphi_i(v_j) = 1$ when $i = j$ and $\varphi_i(v_j) = 0$ otherwise. It turns out that β' is a basis for V', but one must check that its linearly independent and look at the dimension.

Theorem 3 (Decomposition by Dual Basis)**.** Given a basis $\beta = \{v_1, \ldots, v_n\}$ of V and its corresponding dual basis $\beta' = \{\varphi_1, \ldots, \varphi_n\}$ we can write any linear functional $\varphi \in V'$ as the linear combination

$$\varphi = \varphi(v_1)\varphi_1 + \cdots + \varphi(v_n)\varphi_n.$$

Definition 4 (Dual Map)**.** For any linear map $T : V \to W$ we consider what is called its **dual map** $T' : W' \to V'$ defined by $T'(\psi) = \psi \circ T$ where $\psi \in W'$. Observe that $\psi \circ T : V \to \mathbb{F}$ goes from V to $\mathbb{F}$ and so is indeed an element of V'.

Theorem 4 (Algebraic Properties of Dual Maps)**.**

$$\text{(i) } (S+T)' = S' + T' \quad \text{(ii) } (\lambda T)' = \lambda T' \quad \text{(iii) } (S \circ T)' = T' \circ S'$$

Definition 5 (Annihilator)**.** For any subspace U of a space V we can define a subspace

$$U^0 = \{\varphi \in V' | \varphi(u) = 0 \, \forall u \in U\}$$

of V' called the **annihilator of U** which consists of dual vectors in V' which send everything in U to 0.

Proposition 1 (Dimension of Annihilator)**.** $\dim U^0 = \dim V - \dim U$. Same as V/U? How interesting. See Problem 2.

Theorem 5 (Dual Maps and Annihilators)**.** For any map $T : V \to W$ between finite dimensional spaces we have

$$\text{(i) } \operatorname{Null} T' = (\operatorname{Range} T)^0 \quad \text{and} \quad \text{(ii) } \operatorname{Range} T' = (\operatorname{Null} T)^0.$$

You only have to remember these two facts and the dimension of the annihilator to derive the rest of the formulas for $\dim \operatorname{Null} T'$ and $\dim \operatorname{Range} T'$.

Theorem 6 (Polynomial Facts)**.**

(i) For any two polynomials p, s we can find polynomials q, r with $\deg r < \deg s$ such that $p = sq + r$.

(ii) For a polynomial p with complex coefficients we can always write $p(z) = (z-\lambda_1)\cdots(z-\lambda_n)$ as product of linear factors. The complex numbers $\lambda_1, \ldots, \lambda_n$ are precisely the roots of p and may repeat.

(iii) For a polynomial p with real coefficients the best we can do is $p(x) = (x-\lambda_1)\cdots(x-\lambda_k)(x^2+b_1x+c_1)\cdots(x^2+b_lx+c_l)$ with some linear and some quadratic factors. The real numbers $\lambda_1, \ldots, \lambda_k$ are precisely the real roots of p.

(iv) We can multiply polynomials $(pq)(z) = p(z)q(z)$.

Definition 6 (Invariant Subspace)**.** For a linear transformation $T : V \to V$, often called an **operator on V** to call attention to the fact its codomain is also V, if U is a subspace of V such that $Tu \in U$ for all $u \in U$ then we say that U is an **invariant subspace** of T.

Definition 7 (Minimal Polynomial)**.** For any operator $T : V \to V$ we can look at polynomials of T of the form

$$p(T) = a_0 I + a_1 T + \cdots a_k T^k$$

where $I = id_V$ and T^i is the composition of T with itself i times. The **minimal polynomial** of T is the unique smallest degree polynomial m_T such that $m_T(T) = 0$ with coefficient on the highest degree equal to 1.

Definition 8 (Eigenpairs)**.** $\lambda \in \mathbb{F}$ is an **eigenvalue** of $T \in \mathcal{L}(V)$ if there's a nonzero **eigenvector** $v \in V$ with $T(v) = \lambda v$.

Definition 9 (Eigenspace)**.** Eigenspace $E(\lambda, T) := \{v \in V | Tv = \lambda v\}$ is the set of all eigenvectors associated with an eigenvalue λ of T.

Proposition 2. For $T \in \mathcal{L}(V), \lambda \in \mathbb{F}$, the following are equivalent:

(i) λ is an eigenvalue of T (ii) $T - \lambda I$ is not invertible (iii) $T - \lambda I$ is not injective (iv) $T - \lambda I$ is not surjective.

Proposition 3. Any eigenvectors $v_1, \ldots, v_n$ corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ are linearly independent.

Corollary 1. If V is finite-dimensional, any operator $T \in \mathcal{L}(V)$ has at most $\dim V$ distinct eigenvalues.

Theorem 7 (Eigenvalues and Minimal Polynomial)**.** For any operator $T : V \to V$ the zeros of its minimal polynomial m_T are precisely the eigenvalues of T.

Definition 10 (Upper Triangular and Diagonal Matrix)**.** A square matrix $\mathcal{M}$ with coefficients in some field $\mathbb{F}$ is called **upper triangular** if all coefficients below the diagonal are 0. If *all* coefficients off the diagonal are 0 we say that M is **diagonal**.

$$\begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix} \qquad \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix}$$

Definition 11. T is **diagonalizable** if V has a basis $\beta = \{v_i\}$ so $M_\beta(T)$ is diagonal.

Proposition 4. Fix a linear map $T\colon V \to V$ of vector spaces. Then the following are equivalent.

(i) T is diagonalizable (ii) V has a basis of T-eigenvectors (iii) $V = \oplus_{i=1}^m E(\lambda_i, T)$ for distinct $\{\lambda_i\}_1^m$

Proposition 5. $T \in \mathcal{L}(V)$ is diagonalizable iff T's minimal polynomial splits like $(x-\lambda_1)\cdots(x-\lambda_n)$ for distinct λ_i.

Theorem 8 (Eigenvalues of Upper Triangular)**.** If $T : V \to V$ is an operator such that with respect to some basis β of V the matrix $\mathcal{M}_{\beta,\beta}(T)$ is upper triangular, then the coefficients $\lambda_1, \ldots, \lambda_n$ on the diagonal (some may repeat) are precisely the eigenvalues of T.

Exercises

Problem 1 (Dual Space Practice)**.** Consider the vector real space $V := \operatorname{span}\{1, \sin x, \sin 2x\}$ as a subspace of $C([-\pi, \pi], \mathbb{R})$, the vector space continuous real valued functions from $[-\pi, \pi]$ to $\mathbb{R}$.

(a) Show that the linear functionals $\delta_0(f) = f(0), \delta_{\pi/2}(f) = f(\pi/2)$, and $\delta_0'(f) = \frac{df}{dx}(0)$ span, and therefor form a basis of, the dual space of V.

(b) By definition, the vectors $\{1, \sin x, \sin 2x\}$ form a basis for V. Find this list's corresponding dual basis $\{\varphi_1, \varphi_2, \varphi_3\}$ written as linear combinations of $\delta_0, \delta_{\pi/2}$, and δ_0'.

(c) Consider the linear functional $\varphi(f) = \frac{1}{\pi}\int_0^\pi f(x)\,dx$ on V. Find φ's linear combination in the basis $\{\varphi_1, \varphi_2, \varphi_3\}$ and then use this to write φ as a linear combination of the dual vectors $\delta_0, \delta_{\pi/2}$, and δ_0'.

Problem 2. Let V be a finite dimensional vector space and U a subspace of V. We have maps $i : U \to V$, $i(u) = u$ and $\pi : V \to V/U$, $\pi(v) = v + U$ which are injective and surjective respectively.

(a) Calculate $\operatorname{Null} i'$, $\operatorname{Range} i'$, $\operatorname{Null} \pi'$, and $\operatorname{Range} \pi'$ where $i' : V' \to U'$ and $\pi' : (V/U)' \to V'$ are the corresponding dual maps. (By calculate I mean express as some annihilator or other subspace).

(b) Consider the quotient space $V'/\operatorname{Range} \pi'$ and construct a map $T : U' \to V'/\operatorname{Range} \pi'$ and prove that it is an isomorphism.

Problem 3. For $p \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$ with eigenpair (λ, v), show $p(T)v = p(\lambda)v$.

Problem 4. Let V denote the $\mathbb{R}$-vector space space spanned by the functions $\{1, \sin x, \cos x, \sin 2x, \cos 2x\}$, and let $d\colon V \to V$ denote the differentiation operator.

1. What are the eigenvalues of d? Is d diagonalizable?
2. Define the linear map $T\colon \mathbb{C}^2 \to \mathbb{C}^2$ by $T(x, y) \coloneqq (y, -x)$. What are the eigenvalues of T? Find a basis of $\mathbb{C}^2$ such that the matrix associated to T is upper-triangular.
3. Define the linear map $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x, y) \coloneqq (y, -x)$. Show that there is no basis of $\mathbb{R}^2$ such that the matrix associated to T is upper-triangular.

Problem 5. Find the minimal polynomial, the characteristic polynomial, the eigenvalues, and the eigenspaces of the following matrix thought of as an operator $\mathbb{R}^3 \to \mathbb{R}^3$

$$\begin{pmatrix} 3 & 1 & 1 \\ 0 & 3 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

Problem 6 (Axler 5C.7)**.** Suppose $T : V \to V$ is an operator on a finite dimensional vector space V and $v \in V$.

(a) Prove that there exists a unique smallest degree polynomial p_v with leading coefficient 1 such that $p_v(T)v = 0$

(b) Prove that the minimal polynomial of T is a multiple of p_v. (This actually gives a practical way of calculating m_T in most cases).

Problem 7. For any operator $T : V \to V$ prove that the following spaces are invariant under T: (i) $\operatorname{Null} T$, (ii) $\operatorname{Range} T$. Furthermore prove that their annihilators are invariant under T'.

Problem 8 (True/False)**.**
(T/F): If v_1, v_2 are eigenvectors of T then so is $v_1 + v_2$
(T/F): If T is injective then so is its dual map T'
(T/F): Any vector space can be written as a direct sum of eigenspaces of some operator.
(T/F): The null space of $T + S$ is always $\operatorname{Null} T + \operatorname{Null} S$.
(T/F): An operator on a real vector space always has an eigenvector.
(T/F): The dimension of the eigenspace $E(\lambda, T)$ is same as the multiplicity of $(z - \lambda)$ in the minimal polynomial.
(T/F): For an upper triangular matrix, the minimal polynomial of this operator depends on the values above the diagonal.